Entanglement cost of every two-qubit state

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Abstract

We establish strong superadditivity of entanglement of formation for arbitrary finite-dimensional mixed states with a distinguished two-qubit subsystem. Consequently, entanglement of formation is additive for every two-qubit state paired with any finite-dimensional bipartite state, and the asymptotic entanglement cost equals the Wootters expression. The proof combines a balanced rectangular block-matrix entropy inequality derived from the Schatten–Hanner inequality with a transverse-correlation bound for concurrence at fixed marginal eigenaxes.

1 Definitions and result

All Hilbert spaces are finite dimensional. We use ordinary tensor products and partial traces, without superselection restrictions. Entropies are in bits except during the differentiation in Section 2. For a bipartite state,

$$E_F(\rho_{A:B}) = \inf_{\rho=\sum_j q_j|\psi_j\rangle\langle\psi_j|} \sum_j q_j S(\mathrm{Tr}_B\,|\psi_j\rangle\langle\psi_j|).$$

Write $h_2(t) = -t\log_2 t - (1-t)\log_2(1-t)$, with $0\log 0 = 0$, and

$$W(c) = h_2\left(\frac{1+\sqrt{1-c^2}}{2}\right), \qquad 0 \le c \le 1.$$

For a two-qubit state let $J = \sigma_y \otimes \sigma_y$, $\widetilde{\rho} = J\rho^* J$, and let $\lambda_1 \ge \cdots \ge \lambda_4$ be the square roots of the eigenvalues of $\rho\widetilde{\rho}$. Wootters' theorem [1] states

$$C(\rho) = \max(0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4), \qquad E_F(\rho) = W(C(\rho)).$$

We also use Wootters' convex-roof characterization of concurrence, which implies convexity, and the monotonicity of W. Product ensembles give subadditivity of E_F.

Theorem. *Let A, B be qubits and let A', B' have arbitrary finite dimensions. Every state $\Omega_{AA'BB'}$ satisfies*

$$\boxed{E_F(\Omega_{AA':BB'}) \ge E_F(\Omega_{A:B}) + E_F(\Omega_{A':B'}).} \tag{1}$$

Consequently, for any two-qubit state ρ and any finite-dimensional bipartite state σ,

$$E_F(\rho\otimes\sigma) = E_F(\rho) + E_F(\sigma). \tag{2}$$

In particular, for all two-qubit states,

$$\boxed{E_F(\rho^{\otimes n}) = nW(C(\rho)) \quad \text{for every } n \ge 1.} \tag{3}$$

The operational entanglement cost under unrestricted LOCC, free classical communication, vanishing trace-distance error and no extra nonlocal resource is therefore

$$\boxed{E_C(\rho) = W(C(\rho)).} \tag{4}$$

The proof combines the balanced block entropy inequality of Lemma 1 with the fixed-axis concurrence bound of Lemma 2. The imported results are the Schatten–Hanner inequality [2], Wootters' formula [1], and the operational regularization theorem [3].

2 Balanced block entropy

Lemma 1. *Let M be a complex $2r$-by-$2s$ matrix of Frobenius norm one, partitioned into r-by-s blocks M_{ij}. Put $p_{ij} = \|M_{ij}\|_2^2$, using the Schatten 2 norm. Suppose*

$$p_{00} = p_{11} = x/2, \qquad p_{01} = p_{10} = y/2, \qquad x + y = 1.$$

Then

$$S(MM^\dagger) \geq W(|x-y|) + \sum_{i,j} p_{ij} S(M_{ij}M_{ij}^\dagger/p_{ij}). \tag{5}$$

A zero-weight term is defined to be zero.

Proof. Heinävaara's Corollary 1.1 [2] gives, for arbitrary complex matrices X, Y and $q \geq 2$,

$$\|X+Y\|_q^q + \|X-Y\|_q^q \leq (\|X\|_q + \|Y\|_q)^q + |\|X\|_q - \|Y\|_q|^q. \tag{6}$$

Corollary 1.5 covers arbitrary complex matrices X, Y; zero-padding includes rectangular matrices while preserving singular values.

Set

$$X = \begin{pmatrix} M_{00} & 0 \\ 0 & M_{11} \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & M_{01} \\ M_{10} & 0 \end{pmatrix}.$$

The matrix $X - Y$ is obtained from $M = X + Y$ by changing the signs of the lower block row and right block column. It has the same singular values as M. Thus (6) becomes

$$2\|M\|_q^q \leq (r(q) + s(q))^q + |r(q) - s(q)|^q, \tag{7}$$

where $r(q) = \|X\|_q$, $s(q) = \|Y\|_q$. Both sides equal 2 at $q = 2$.

Use natural logarithms for the next calculation, writing S_e and $W_e = (\ln 2)W$. For any fixed nonzero matrix Z, with $z = \|Z\|_2^2$, differentiation of its singular-value sum gives

$$\frac{d}{dq} \log \|Z\|_q \bigg|_2 = -\frac{1}{4} S_e(ZZ^\dagger/z). \tag{8}$$

Indeed $\log \|Z\|_q = q^{-1} \log \sum_k \zeta_k^q$; substituting $\zeta_k^2 = z\mu_k$ in its derivative cancels the $\log z$ terms. Zero singular values contribute identically zero.

For $x, y > 0$, let $S_X = S_e(XX^\dagger/x)$, $S_Y = S_e(YY^\dagger/y)$. At $q = 2$, $r = \sqrt{x}$, $s = \sqrt{y}$, $r' = -rS_X/4$, and $s' = -sS_Y/4$. The derivative of the left side of (7) is $-S_e(MM^\dagger)$. The derivative of the right side is

$$\begin{aligned} &(r+s)^2 \log(r+s) + (r-s)^2 \log|r-s| - xS_X - yS_Y \\ &= \ln 2 - W_e(|x-y|) - xS_X - yS_Y. \end{aligned} \tag{9}$$

Equality at 2 in (7) gives the same inequality for the right derivatives. Rearranging yields

$$S_e(MM^\dagger) \geq W_e(|x-y|) + xS_X + yS_Y - \ln 2. \tag{10}$$

Block-diagonal entropy and the assumed weights give

$$xS_X + yS_Y = \ln 2 + \sum_{ij} p_{ij} S_e(M_{ij}M_{ij}^\dagger/p_{ij}). \tag{11}$$

Substitution and division by $\ln 2$ prove (5).

Without the balance assumption, the same argument gives

$$\begin{aligned} S(MM^\dagger) \geq W(|x-y|) + \sum p_{ij} S(M_{ij}M_{ij}^\dagger/p_{ij}) \\ + H(p_{00}, p_{01}, p_{10}, p_{11}) - h_2(x) - 1. \end{aligned} \tag{12}$$

Balance is exactly what cancels the final three classical-entropy terms.

If $x = y$, differentiability of r, s gives $|r(q) - s(q)| = O(|q-2|)$, so the absolute-difference term in (7) has zero right derivative. If $x = 0$ or $y = 0$, M is block diagonal or block antidiagonal and (5) is the direct-sum entropy identity. Individual zero blocks contribute zero by continuity. These endpoint limits are taken at fixed finite dimensions and fixed tensor power. □

3 Transverse spin correlations bound concurrence

Write the Bloch decomposition

$$\rho = \frac{1}{4}\left(I \otimes I + a \cdot \sigma \otimes I + I \otimes b \cdot \sigma + \sum_{ij} T_{ij} \sigma_i \otimes \sigma_j \right).$$

Choose any unit axes n, m for which a is parallel to n and b is parallel to m. If a Bloch vector is zero, its axis may be chosen arbitrarily. Let $U = n^{\perp}$, $V = m^{\perp}$, each a fixed two-dimensional real plane, and define

$$k_{n,m}(\rho) = \max_{u \in U, v \in V,\ |u|=|v|=1} |u^T T v|.$$

Lemma 2. *For every two-qubit state and every such choice of axes,*

$$\boxed{C(\rho) \le k_{n,m}(\rho).} \tag{13}$$

3.1 Rank at most two

Proof. A two-dimensional complex subspace of $\mathbb{C}^2 \otimes \mathbb{C}^2$ contains a product vector: represent two independent vectors as 2-by-2 matrices; the homogeneous quadratic determinant of their linear combination has a projective complex root. The corresponding nonzero matrix has rank one. A rank-at-most-two state therefore has a product vector in its kernel. Local unitaries put that vector at $|01\rangle$.

Positivity makes the 01 row and column vanish. Factor the state as $ZZ^{\dagger}$ with two columns (pad a rank-one factor with a zero column). A unitary on these columns makes the 00 row equal to $(L, 0)$, $L \ge 0$. If $L = 0$, the state is supported on $|1\rangle_A \otimes \mathbb{C}^2_B$, so its concurrence is zero and (13) is immediate. Otherwise the two local relative phases and the remaining second-column phase put the factor in the form

$$Z = \begin{pmatrix} L & 0 \\ 0 & 0 \\ u + iv & s \\ t & r \end{pmatrix}, \qquad L, s, t, r \ge 0, \quad u, v \in \mathbb{R}. \tag{14}$$

The phases of s, t, r change by $\alpha + \gamma$, $\alpha + \beta$, $\alpha + \beta + \gamma$, respectively. These independent combinations make the nonzero entries nonnegative, in agreement with the three-qubit canonical purification form [4].

Temporarily leave the trace unnormalized:

$$N = L^2 + u^2 + v^2 + s^2 + t^2 + r^2, \qquad c = 2Lt.$$

The two potentially nonzero spin-flip roots are the singular values of

$$Z^T J Z = \begin{pmatrix} -c & -Lr \\ -Lr & 0 \end{pmatrix}. \tag{15}$$

This follows by the equality of the nonzero spectra of AB and BA, applied to $ZZ^{\dagger} J \overline{Z} Z^T J$. The singular-value difference of (15) is c. Thus $C(ZZ^{\dagger}/N) = c/N$.

Direct Pauli traces of $ZZ^{\dagger}$ give the unnormalized vectors

$$a = (2Lu, 2Lv, 2L^2 - N), \qquad b = (2(tu + sr), -2tv, N - 2(t^2 + r^2)), \tag{16}$$

and

$$\begin{aligned} T &= \begin{pmatrix} c & 0 & a_x \\ 0 & -c & a_y \\ -b_x & -b_y & 2(L^2 + t^2 + r^2) - N \end{pmatrix} \\ &= D + a e_z^T - e_z b^T, \quad D = \operatorname{diag}(c, -c, N). \end{aligned} \tag{17}$$

For $u_0 \in U$, $v_0 \in V$, both correction terms vanish, even when a Bloch vector is zero:

$$u_0^T T v_0 = u_0^T D v_0. \tag{18}$$

If $c = 0$, the claim is immediate. Otherwise D is invertible, and the two planes V and $D^{-1}U$ in $\mathbb{R}^3$ intersect nontrivially. Pick a unit vector v_0 in their intersection and set $u_0 = Dv_0/\|Dv_0\|$. Since

$$N \geq L^2 + t^2 \geq 2Lt = c,$$

the least singular value of D is c, so

$$u_0^T T v_0 = \|Dv_0\| \geq c. \tag{19}$$

Dividing by N proves the lemma for every rank-at-most-two state. Transporting the prescribed axes through the local rotations preserves every admissible marginal eigenaxis choice.

3.2 Extension to all ranks

Work in the chosen local marginal eigenbases, and set $w = \rho_{00,11}$, $z = \rho_{01,10}$. Direct evaluation of the 2-by-2 transverse correlation matrix gives

$$k_{n,m}(\rho) = 2(|w| + |z|). \tag{20}$$

Independent local diagonal phases make $w, z \geq 0$ and act as orthogonal rotations within U, V. In this gauge the transverse matrix is $\mathrm{diag}(2(w+z), 2(z-w))$.

Consider the compact convex set $\mathcal{F}$ of density matrices whose local marginals are diagonal in these fixed bases, whose w, z are real, and for which $w, z \geq 0$. There are seven real affine equality constraints: trace one, four local off-diagonal marginal conditions, and two imaginary anti-diagonal conditions.

Suppose an extreme point in $\mathcal{F}$ has rank $q \geq 3$ and at least one of w, z positive. Hermitian perturbations supported on its range have $q^2 \geq 9$ real parameters. At most one nonnegativity constraint is active. Thus at most eight homogeneous linear conditions constrain such perturbations, leaving a nonzero perturbation H. For sufficiently small $\epsilon > 0$, both $\rho \pm \epsilon H$ remain positive on the support, preserve all equalities and active constraints, and preserve the inactive strict signs. They are distinct points of $\mathcal{F}$, contradicting extremality. Therefore every extreme point has rank at most two or satisfies $w = z = 0$.

On the zero face, diagonal marginals give

$$\rho = \begin{pmatrix} \alpha & x & y & 0 \\ x^* & \beta & 0 & -y \\ y^* & 0 & \gamma & -x \\ 0 & -y^* & -x^* & \delta \end{pmatrix}. \tag{21}$$

Independent local diagonal phases make x, y real. For a real state, $R = \sqrt{\rho} J \sqrt{\rho}$ is real symmetric and the Wootters roots are the absolute values of its eigenvalues μ_j, because $\sqrt{\rho}\tilde{\rho}\sqrt{\rho} = R^2$. The triangle inequality gives

$$C(\rho) = \max(0, 2\max_j |\mu_j| - \sum_j |\mu_j|) \leq \left| \sum_j \mu_j \right| = |\operatorname{Tr}(\rho J)| = 0. \tag{22}$$

Thus the zero face has zero concurrence. The same separability also follows from the two-qubit PPT criterion.

Every member of a compact finite-dimensional convex set is a convex combination of its extreme points. On $\mathcal{F}$, concurrence is convex and $2(w+z)$ is linear. Applying the rank-at-most-two bound and the zero-face result to such a decomposition gives

$$C(\rho) \leq \sum_j p_j C(\rho_j) \leq \sum_j 2p_j(w_j + z_j) = 2(w+z) = k_{n,m}(\rho).$$

Lemma 2 follows at every rank, including degenerate marginal spectra. □

4 Closure to strong superadditivity and operational cost

Proof of the theorem. Let $|\Psi\rangle$ be any pure state of $AA'BB'$, and let ρ_{AB} be its two-qubit marginal. Choose axes n, m as in Lemma 2. The unit spheres in the two planes are compact, so there are maximizing transverse vectors u, v. Reversing one sign if necessary gives

$$\kappa = u^T T v = k_{n,m}(\rho_{AB}) \geq C(\rho_{AB}) \geq 0.$$

Expand the qubits in the eigenbases of $u \cdot \sigma$ and $v \cdot \sigma$. Because $u \cdot a = v \cdot b = 0$, the local measurement distributions are both uniform. The four joint probabilities are exactly

$$p_{00} = p_{11} = \frac{1+\kappa}{4}, \qquad p_{01} = p_{10} = \frac{1-\kappa}{4}. \tag{23}$$

Across $AA' : BB'$, the coefficient matrix of $|\Psi\rangle$ has four A'-by-B' blocks with these squared norms. Let $|\phi_{ij}\rangle$ be the corresponding normalized nonzero block vectors. Lemma 1 and monotonicity of W give

$$\begin{aligned} S(\Psi_{AA'}) &\geq W(\kappa) + \sum_{ij} p_{ij} S(\phi_{ij,A'}) \\ &\geq E_F(\rho_{AB}) + \sum_{ij} p_{ij} E(\phi_{ij}). \end{aligned} \tag{24}$$

Tracing out AB shows

$$\rho_{A'B'} = \sum_{ij} p_{ij} |\phi_{ij}\rangle\langle\phi_{ij}|.$$

The definition of the convex roof therefore yields

$$S(\Psi_{AA'}) \geq E_F(\rho_{AB}) + E_F(\rho_{A'B'}). \tag{25}$$

Take any pure decomposition $\Omega = \sum_\ell q_\ell |\Psi_\ell\rangle\langle\Psi_\ell|$. Apply (25) to each vector. The measurement axes may depend on ℓ, since the right-hand side contains only intrinsic marginal entanglement measures. Convexity gives

$$\sum_\ell q_\ell S(\Psi_{\ell,AA'}) \geq \sum_\ell q_\ell \{E_F(\rho^\ell_{AB}) + E_F(\rho^\ell_{A'B'})\} \geq E_F(\Omega_{AB}) + E_F(\Omega_{A'B'}).$$

Taking the infimum over all decompositions, without assuming attainment, proves (1).

For $\Omega = \rho \otimes \sigma$, (1) gives the lower bound in (2); products of arbitrarily close-to-optimal ensembles give the upper bound. Iteration proves (3) for every finite n, including boundary states directly. The operational regularization theorem [3],

$$E_C(\rho) = \lim_{n\to\infty} \frac{E_F(\rho^{\otimes n})}{n},$$

then gives (4). The theorem in [3] uses Bures error; dimension-independent fidelity/trace-distance inequalities identify its vanishing-error criterion with the trace-distance criterion used here. □

5 Scope and physical meaning

Equation (4) identifies the asymptotic Bell-pair formation rate of an ordinary two-qubit mixed state. Equation (3) shows that collective decompositions leave the per-copy convex-roof cost unchanged. Reversibility with entanglement distillation is a separate question.

The distinguished marginal in (1) is 2×2, with arbitrary auxiliary dimensions. The additivity conclusion allows any finite-dimensional partner of a two-qubit state. Iterating (1) over qubit pairs bounds entanglement by a sum of pair-marginal entanglements, which can be strictly smaller than the entanglement of a grouped marginal. For crossed Bell pairs A_1B_2 and A_2B_1, the grouped formation entanglement is 2 bits while both same-index marginals have zero entanglement.

Fermionic superselection uses a different convex roof and copy-composition convention. With ordinary-state conventions, the example in [7, Eq. III.28] is separable.

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